

**Name: Aarya Katulwar**

## **Project Title: AI-Powered Resume Ranker**

### **Abstract**

The project automates resume shortlisting by ranking candidate resumes against a job description. Using TF-IDF vectorization, cosine similarity, skill extraction, and experience heuristics, the system simulates a lightweight Applicant Tracking System (ATS). It outputs ranked candidates in a CSV format, helping recruiters identify the most relevant resumes quickly.

### **Tools Used**

- Python
- scikit-learn (TF-IDF, cosine similarity)
- pdfminer.six (PDF text extraction)
- pandas, numpy (data handling)
- regex (skills & experience extraction)
- Streamlit (optional, for UI demo)

### **Steps Involved**

- Collected job description text and sample resumes (in TXT/PDF format).
- Performed text extraction and preprocessing (cleaning, normalization).
- Applied TF-IDF Vectorization to convert text into numerical vectors.
- Computed Cosine Similarity between JD and resumes.
- Extracted skills from resumes using regex and dictionary matching.
- Extracted years of experience using regex patterns.
- Designed weighted scoring (60% similarity, 30% skills, 10% experience).
- Ranked resumes and saved results to CSV.

### **Results**

The system successfully ranked resumes against the job description. Top resumes matched over 80% of required skills. The ranked output includes candidate name, similarity score, skills matched, experience years, and final score.

## Conclusion

This project demonstrates how AI and NLP techniques can be applied to HR automation. The AI-Powered Resume Ranker provides a simple yet effective approach for resume shortlisting. Future improvements include using semantic embeddings (e.g., Sentence-BERT) and advanced NER-based skill extraction for better accuracy.